b) $\sigma_k = \int_a^b l_k(x)\,dx$

$[a, b]$

$x_0 \quad x_2$

$\sigma_2 = \int_a^b l_2(x)\,dx$

$x_1 = \frac{a+b}{2} = m$

$$l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

$$l_2(x) = \frac{(x-a)(x-m)}{(b-a)(b-m)}$$

$$\sigma_2 = \int_a^b \frac{(x-a)(x-m)}{(b-a)(b-m)}\,dx$$

$$= \frac{1}{(b-a)(b-m)} \int_a^b (x-a)(x-m)\,dx$$

$$= \frac{1}{(b-a)(b-m)} \int_a^b (x^2 - xm - ax + am)\,dx$$

$$= \frac{1}{(b-a)(b-m)} \left[\frac{x^3}{3} - \frac{x^2 m}{2} - \frac{ax^2}{2} + amx \right]_a^b$$

$$= \frac{1}{(b-a)(b-m)} \left[\left(\frac{b^3}{3} - \frac{b^2 m}{2} - \frac{ab^2}{2} + amb \right) - \left(\frac{a^3}{3} - \frac{a^2 m}{2} - \frac{a^3}{2} + a^2 m \right) \right]$$

For derivation, showing till this much will be sufficient you can skip the steps for further simplification.

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$$= \frac{1}{(b-a)(b-m)}\left(\frac{b^3}{3} - \frac{b^2 m}{2} - \frac{ab^2}{2} + amb - \frac{a^3}{3} + \frac{a^2 m}{2} + \frac{a^3}{2} - a^2 m\right)$$

$$= \frac{1}{(b-a)\left(b-\frac{a+b}{2}\right)}\left\{\frac{b^3}{3} - \frac{b^2}{2}\times\frac{(a+b)}{2} - \frac{ab^2}{2} + ab\times\frac{(a+b)}{2} - \frac{a^3}{3} + \frac{a^2}{2}\times\frac{(a+b)}{2} + \frac{a^3}{2} - a^2\left(\frac{a+b}{2}\right)\right\}$$

$$= \frac{1}{(b-a)\left(\frac{b-a}{2}\right)}\left(\frac{b^3}{3} - \frac{ab^2}{4} - \frac{b^3}{4} - \frac{ab^2}{2} + \frac{a^2 b}{2} + \frac{ab^2}{2} - \frac{a^3}{3} + \frac{a^3}{4} + \frac{a^2 b}{4} + \frac{a^3}{2} - \frac{a^3}{2} - \frac{a^2 b}{2}\right)$$

$$= \frac{2}{(b-a)^2}\left(\frac{b^3}{12} - \frac{3ab^2}{4} + \frac{3a^2 b}{4} + \frac{ab^2}{2} - \frac{a^3}{12} - \frac{a^2 b}{2}\right)$$

$$= \frac{2}{(b-a)^2}\left(\frac{b^3}{12} - \frac{ab^2}{4} + \frac{a^2 b}{4} - \frac{a^3}{12}\right)$$

$$= \frac{2}{(b-a)^2} \times \frac{b^3 - 3ab^2 + 3a^2 b - a^3}{12}$$

$$= \frac{2}{(b-a)^2} \times \frac{(b-a)^3}{12}$$

$$= \frac{b-a}{6}$$

$$\therefore \sigma_2 = \frac{1}{6}(b-a) \quad \text{[verified]}$$

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